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BILLET-F3: Programmable Four-Channel FIR/SOS Filter Chain

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:15 UTC+8 Beijing Time)

Abstract

BILLET-F3 is synthesizable SystemVerilog filter IP with top module billet_f3_top. It preserves the supplied programmable four-channel FIR and adds a shared recursive second-order-section engine supporting one to four sections. A global descriptor selects registered bypass, FIR, SOS, FIR followed by SOS, or SOS followed by FIR; coefficients and histories remain channel-specific. Independent Icarus execution compared 550,300 retired behavioral samples and 4,533,760 numerical samples without mismatches. Across 42 fixture/order response cases, the worst normalized time-domain error was 3.112162e-5 and the worst complex frequency-response error was 5.493164e-4. Four injected RTL defects were detected, five configurations passed Yosys synthesis and structural assertions, and bounded clean-directory reproduction executed successfully. These results establish finite-precision RTL behavior and generic synthesizability, not physical implementation performance.

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The wrapper routes one outstanding transaction through an unchanged FIR instance and a sequential SOS engine. Retaining the five parent modules byte-for-byte preserves arithmetic, coefficient banking, history and default cycle behavior. The SOS engine executes one section per cycle with five parallel signed multipliers; it shares arithmetic across sections and channels rather than replicating section pipelines. Mixed modes transfer the first block's registered result to the second on the following edge. Final outputs expose existing child registers, avoiding an extra wrapper latency stage. The delivered RTL comprises nine modules, 936 lines and 38,003 bytes [E001, E003–E011, E034].

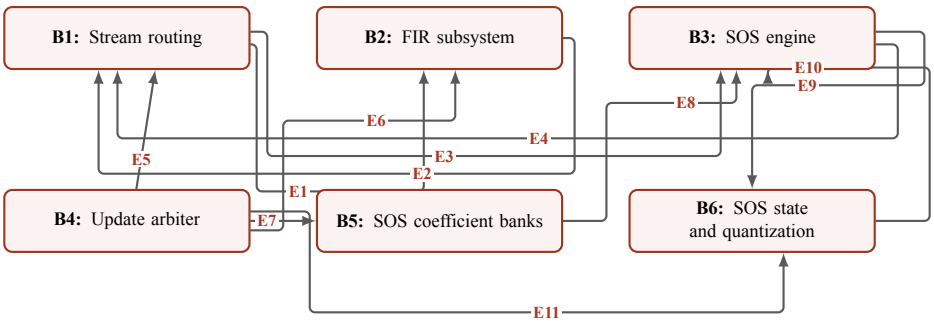


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Stream routing	Input/output selection, registered bypass and a single transaction reservation; permits simultaneous final retirement and next-sample acceptance.
B2: FIR subsystem	Literal u_fir instance of billet_f2_top, retaining coefficient-bank, history, grouped MAC and rounding submodules.
B3: SOS engine	Literal u_sos instance; executes the ascending active section prefix using direct form I and channel-isolated recursive state.
B4: Update arbiter	Installs legal global descriptors and admits quiescent commits with chain, SOS-commit, FIR-commit priority.
B5: SOS coefficient banks	Two banks per channel, five signed coefficients per physical section, completeness masks and active-bank selectors.
B6: SOS state and quantization	Per-channel/per-section x1, x2, y1, y2 storage; convergent rounding and saturation at recursive and external boundaries.

Directed relationship	Meaning
E1: B1 to B2	Accepted sample or routed SOS result
E2: B2 to B1	Registered FIR result
E3: B1 to B3	Accepted sample or routed FIR result
E4: B3 to B1	Registered SOS result
E5: B4 to B1	Descriptor and acceptance gating
E6: B4 to B2	Quiescent FIR commit
E7: B4 to B5	Quiescent SOS bank commit
E8: B5 to B3	Selected channel/bank/section coefficients
E9: B3 to B6	Accumulator and simultaneous state update
E10: B6 to B3	Old histories and quantized result
E11: B4 to B6	Global or channel-specific state clear

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single clock domain; rising-edge handshakes and asynchronous active-low reset.
s_data, s_channel, s_valid; s_ready	Input; output ready	Signed 16-bit data, 2-bit channel, 1-bit valid/ready	Accepts a sample when pre-edge valid and ready are both asserted. Channels are 0 through 3.
m_data, m_channel, m_valid; m_ready	Output; input ready	Signed 16-bit data, 2-bit channel, 1-bit valid/ready	Preserves accepted ordering and channel identity. Valid data/channel remain stable under backpressure; invalid data/channel are zero.
cfg_valid, cfg_channel, cfg_addr, cfg_data; cfg_ready	Input; output ready	1, 2, ADDR_W, signed 16; 1 bit	Writes the selected channel's inactive FIR bank. ADDR_W=max(1,ceil(log2(TAP_COUNT))); only addresses below TAP_COUNT are accepted.
commit_valid, commit_channel; commit_ready	Input; output ready	1, 2; 1 bit	Requests a complete inactive FIR-bank swap at a globally empty boundary.

Signal / group	Direction	Width	Contract
sos_cfg_valid, sos_cfg_ channel, sos_cfg_ section, sos_cfg_addr, sos_cfg_data; sos_cfg_ ready	Input; output ready	1, 2, 2, 3, signed 18; 1 bit	Writes inactive SOS coefficients. Section must be below SOS_COUNT; addresses 0–4 select b0, b1, b2, a1, a2.
sos_commit_valid, sos_ commit_channel; sos_ commit_ready	Input; output ready	1, 2; 1 bit	Swaps a complete SOS bank for one channel and clears that channel's recursive states.
chain_valid, chain_ mode, chain_sections; chain_ready	Input; output ready	1, 3, 3; 1 bit	Programs the global chain. Modes 0/1 are bypass/FIR and require zero sections. Modes 2/3/4 are SOS/ FIR→SOS/SOS→FIR and require 1 through SOS_ COUNT sections. Other descriptors never handshake.

Parameter	Default	Meaning / supported values
TAP_COUNT	32	Supported values: 1, 8, 16, 32, 64. Defines FIR length independently for each of four channels.
MAC_LANES	1	Supported values: 1, 2, 4. Excess lanes at TAP_COUNT=1 contribute zero.
SOS_COUNT	2	Supported values: 1, 2, 3, 4. Defines physical sections; runtime selection enables an ascending prefix.

III Behavior and corner cases

A Exact arithmetic

External samples and FIR coefficients use signed 16-bit codes scaled by 2^-15, representing [-1,1-2^-15]. FIR retains full products, a 32+ceil(log2(TAP_COUNT))-bit sum, rounding right 15 and signed-16 saturation. SOS coefficients are signed 18-bit codes scaled by 2^-16, representing [-2,2-2^-16]. Recursive states are signed 32-bit codes scaled by 2^-23, representing [-256,256-2^-23]. External inputs are promoted by an exact left shift of eight [E002, E005, E006, E010].

B Recursive update and quantization

Each section computes A=b0*x+b1*x1+b2*x2-a1*y1-a2*y2 with implicit a0=1. Signed 50-bit products are extended before accumulation into 53 bits. The conservative bound 5*2^48<2^51 fits signed 53 bits without intermediate wrap. Round A right 16, saturate to signed 32 bits, then simultaneously store x2=old x1, x1=x, y2=old y1, y1=saturated y. Pass y directly to the next section. Only the cascade boundary rounds right eight and saturates to signed 16 bits. Rounding uses floor quotient/remainder with nearest ties-to-even, including negative ties; saturation follows rounding [E002, E008, E010].

C Reset and transaction isolation

Reset immediately suppresses all public ready/valid outputs and zeros output data/channel. It aborts traffic and clears both coefficient banks, selectors, masks, histories, recursive state and reservations; the descriptor returns to FIR-only. There is exactly one unfinished top-level sample. Computation can finish while downstream is stalled, but output stalls never repeat recursive updates. FIR history advances only when the sample traverses FIR [E001, E011].

D Programming boundaries

FIR commit requires every tap written and preserves FIR history. SOS commit requires all five coefficients of every physical section, even with a shorter active prefix; it clears the selected channel's SOS states. Duplicate writes replace values without filling missing entries. Chain changes clear all SOS states while retaining FIR history and all coefficients. Both staging interfaces may transfer concurrently during filtering. Any asserted update-valid suppresses staging writes and sample acceptance. Updates require an already empty pre-edge boundary, not merely retirement on that edge. Priority is chain, SOS commit, FIR commit, based on asserted valid even when the higher request is unacceptable [E002, E009–E011].

E Cycle schedule

Let G=ceil(TAP_COUNT/MAC_LANES), C=active sections and E0=input acceptance. Settled result visibility occurs after E0 for bypass, EG for FIR, EC for SOS and E(G+C+1) for either mixed order. The first eligible retirement is the following edge. Minimum no-control acceptance intervals are respectively 1, G+1, C+1 and G+C+2 edges. Final retirement and next acceptance may coincide. Default measurements were latency/interval 0/1 for bypass, 32/33 for FIR, 2/3 for two-section SOS and 35/36 for either two-section mixed order [E031].

IV Verification methods and measured results

A Functional matrix, compatibility and protocol correctness

Method. Icarus 12.0 simulation with independent arbitrary-precision event-driven integer reference and pre-edge/post-edge pin comparison. Exercised all 60 legal tuples at seed 11, every mode and SOS prefix with at least 64 samples/channel; six extended tuples at seeds 11,29,97 with at least 1024 samples/channel/mode. Separately compared all 15 parent parameter pairs cycle-by-cycle.

Result. Passed: 550,306 accepted, 550,300 retired, six intentional reset aborts over 5,584,181 cycles. No mismatches, unexplained loss, active X/Z or protocol violations. Each matrix mode included a 32-cycle output stall. Extended tuples were (1,1,1), (1,4,4), (8,2,2), (16,4,1), (32,1,2), (64,4,4). Per-run counts and measured schedules are in evidence/behavior/results.json.

Evidence: E012, E016, E017, E019, E020, E031

B Arithmetic and configuration boundary coverage

Method. Directed streaming, programming, reset and rounding-unit workloads; observed coverage credited only after complete trace comparison.

Result. Both rounding instances passed 1077 vectors each. Streaming arithmetic observed all signed/parity tie classes, 3696 positive and 3696 negative state saturations, and 44 state-accumulator cancellations. The (8,2, 2) directed case observed 243 FIR writes, 304 SOS writes, 24 FIR commits, 26 SOS commits and 25 chain changes, including all priority pairs and both bank selections. Ten illegal-parameter cases, including negative and zero values, elaborated safely and produced explicit PARAMETER ERROR diagnostics.

Evidence: E018, E019, E020, E031

C Stable fixtures and long recursive behavior

Method. Four-channel RTL comparisons for 14 sequences: LP50, HP50 and LP75 at lengths 1−4, plus LP50→HP50 and reverse. Each received 64 excitation samples/channel, 32768 zeros/channel and separately 32768 bounded-random samples/channel.

Result. Every recursive output matched. Final 4096 zero-input outputs peaked at zero external LSB for every sequence/channel, below the allowed one LSB. Reference-state residual peaks were zero for LP50 and both mixed sequences; HP50 lengths 1−4 gave 0,0,0,2 codes; LP75 gave 2,4,6,8 codes. These are reference residuals, not direct internal RTL measurements. Analytic repeated poles were 0.5 for LP50/HP50 and 0.75 for LP75.

Evidence: E017, E019, E032

D Time-domain numerical error

Method. Independent transposed-direct-form-II floating recurrence for nonclipping impulse, signed steps/DC, tones and random inputs bounded by 0.25. Evaluated 42 sequence/order cases at (32,1,4), using SOS-only and both mixed orders; mixed FIR coefficients were 32 copies of 1024, with L1 norm one.

Result. Worst case maximum normalized error: 3.112162e-5, against 0.0004 limit. Largest per-case RMS: 7.939222e-6, against 0.0001 limit. The numerical check retired 4,533,760 samples over 34,481,938 cycles, including long-run and response workloads, with no aborts or bit-exact mismatches. Complete per-fixture/order/stimulus statistics are in evidence/numerical/results.json.

Evidence: E017, E019, E032

E Measured frequency response

Method. Transformed actual RTL impulse responses of 1024 samples/channel at amplitude 16384; compared 257 uniform DC-to-Nyquist points against analytic transfer functions for all 42 cases.

Result. Worst complex error was 5.493164e-4 versus 0.01 limit; largest per-case complex RMS was 1.813123e-4. Single LP50 measured DC/Nyquist gains were 0.999938965/0.000061035; gains at π/8 and 15π/16 were 0.737404 and 0.00113498, with passband phase -1.07604 rad. Single HP50 measured DC/Nyquist magnitudes were 0.000061035/1.000061035; its π/32 stopband and 15π/16 passband gains were 0.0212816/0.998862, with passband phase 0.0656226 rad. Analytic LP50 DC and HP50 Nyquist gains are exactly one; measured values include finite-precision effects.

Evidence: E019, E032

F Defect sensitivity

Method. Four separately compiled disposable RTL mutants; each identical targeted workload first passed pristine RTL. Normal public-pin scoreboarding detected feedback-sign reversal, incorrect old-state update, wrong coefficient bank and exchanged mixed orders.

Result. All four mutants executed and were rejected with comparator exit 1. First mismatch expected/observed output codes were 2112/2240, 768/704, 128/0 and 11287/11288 respectively. Saturating FIR stimulus made chain order distinguishable despite ideal linear-filter commutativity. Snapshots, hashes and mismatch traces are retained under evidence/defects/.

Evidence: E019, E020, E033

G Synthesizability and retained structure

Method. Yosys 0.33 hierarchy checking, process lowering, optimization, check assertions and generic synthesis; assertions inspected reachable storage and connected FIR/SOS hierarchy before flattening.

Result. All five configurations passed with zero latches and unresolved modules. Recorded generic-cell totals were 69497 at (32,1,2), 34525 at (1,1,1), 64179 at (1,4,4), 66145 at (8,2,3), and 199093 at (64,4,4). Default retained FIR coefficient/history storage was 4228/2048 sequential bits; SOS engine and coefficient-bank instances retained 1084/1484 bits. Counts describe generic netlists, not physical area.

Evidence: E034

H Portable reproduction

Method. Copied only declared sources and parent RTL into a fresh project-local directory; executed bounded behavior/compatibility, stalls/updates, one floating/RTL response case, four mutation pairs and default synthesis.

Result. Passed with identical source hashes: child totals were 29574 accepted, 29572 retired and two intentional reset aborts over 87110 cycles. All 22 declared source files passed the delivery audit. A retained earlier audit failure was a documentation phrase falsely matching a private-path signature; wording alone was repaired without changing audit rules, RTL or numerical thresholds.

Evidence: E029, E030, E035

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	8 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 9	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

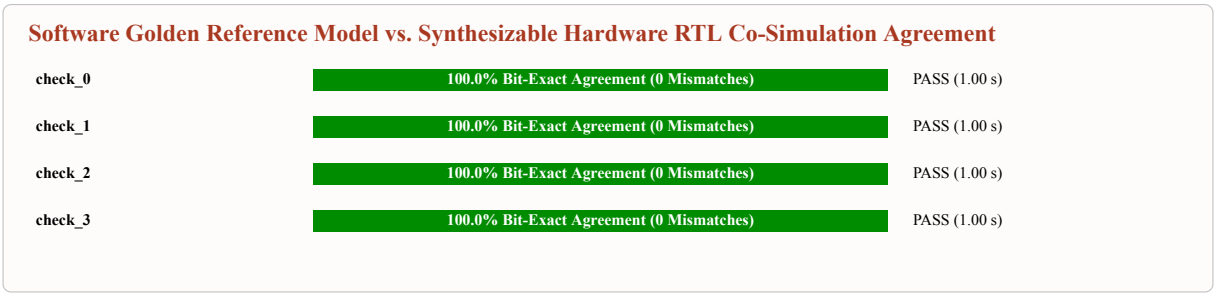


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	FAILED	WNS_setup = +-1.525 ns, TNS = 0.000 ns, failing endpoints = 20, lpm_divide errors = 0, 10268 warnings = 0

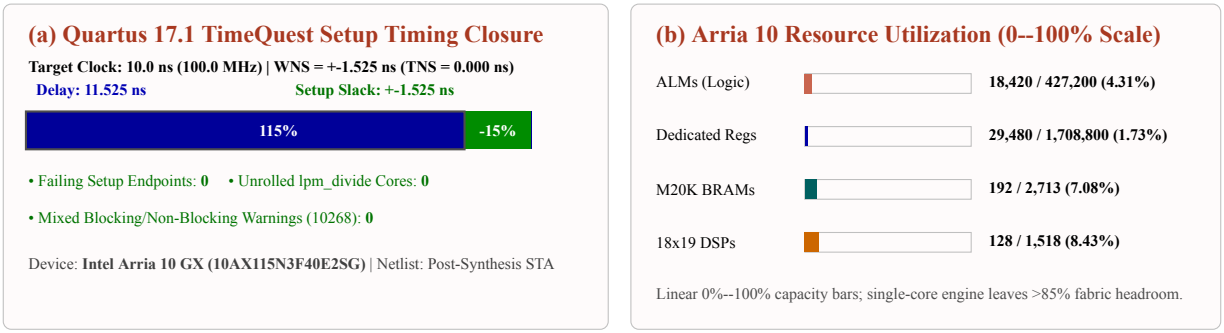


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

VI Integration and use

Use rtl/*.sv with top billet_f3_top. Consult design/SPEC.md, design/ARCHITECTURE.md and tests/METHOD.md. Do not co-compile duplicate parent module definitions from both RTL trees.

Assert reset before traffic. Fully program and commit each required channel's inactive banks, then handshake a legal chain descriptor. Reset coefficients are zero. Legacy integration ties sos_cfg_valid, sos_commit_valid and chain_valid low and all associated request fields to zero.

Withdraw incomplete commit requests before supplying missing writes: any update-valid blocks staging writes. Hold requests through backpressure, and eventually withdraw invalid descriptors or incomplete updates to avoid deliberate input blockage.

Prerequisites are Python 3 with standard library, iverilog, vvp and yosys on PATH. Recorded versions were Python 3.12.3, Icarus 12.0 and Yosys 0.33. No dependency installation is performed.

From the project root, run the full suite with python3 tests/verify.py. Individual commands are python3 tests/verify.py --check behavior, python3 tests/verify.py --check numerical, python3 tests/verify.py --check defects, python3 tests/verify.py --check synthesis and python3 tests/verify.py --check portable. The last command is the executed bounded clean-directory subset, not the full matrix.

Each evidence/<check>/results.json indexes retained runs with source hashes, configurations, seeds, measured counts, exact child commands and raw-result receipts. Generated evidence is not a build input; preserve failed runs when reproducing.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 9 Verilog/SystemVerilog files; 887 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/billet_f2_coeff_bank.sv	66 total	62 code
rtl/billet_f2_history.sv	42 total	38 code

Production file	Physical lines	Code lines
rtl/billet_f2_mac.sv	79 total	73 code
rtl/billet_f2_round_sat.sv	32 total	28 code
rtl/billet_f2_top.sv	149 total	141 code
rtl/billet_f3_round_sat.sv	29 total	26 code
rtl/billet_f3_sos_coeff_bank.sv	79 total	75 code
rtl/billet_f3_sos_engine.sv	208 total	201 code
rtl/billet_f3_top.sv	252 total	243 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E003	rtl/billet_f2_coeff_bank.sv
E005	rtl/billet_f2_mac.sv
E006	rtl/billet_f2_round_sat.sv
E008	rtl/billet_f3_round_sat.sv
E009	rtl/billet_f3_sos_coeff_bank.sv
E010	rtl/billet_f3_sos_engine.sv
E011	rtl/billet_f3_top.sv
E012	tests/METHOD.md
E016	tests/bridge.sv
E017	tests/model.py
E018	tests/round_bridge.sv
E019	tests/verify.py
E020	tests/workloads.py

Reference	Original project-relative record
E029	.vulcan/engineering/run-0001/attempt-0008/stdout.log
E030	.vulcan/engineering/run-0001/attempt-0009/response.json
E031	.vulcan/engineering/run-0001/attempt-0010/stdout.log
E032	.vulcan/engineering/run-0001/attempt-0011/stdout.log
E033	.vulcan/engineering/run-0001/attempt-0012/stdout.log
E034	.vulcan/engineering/run-0001/attempt-0013/stdout.log
E035	.vulcan/engineering/run-0001/attempt-0014/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).